Spyne | Forward Deployed Engineer Case Study

Pune Workshop AI Telephony & Log Diagnostics — 24x7 Real-Time Root Cause Analysis

Candidate: Prateek	**Role:** Forward Deployed Engineer (Full Time)	**Target Site:** Sparsh Motors, Kothrud (KTD-04)
Dataset: 20 Calls + Live Tool Logs	**Submission Date:** October 16, 2026	**Constraints:** ≤ 800 words (a-f), explicit denominators

Executive Summary: Reported Summary vs Verified Ground Truth

Metric	Reported to Management	Audited Ground Truth	Variance / Root Cause
Calls Answered	20 of 20 (100%)	20 of 20 (100%)	Coverage working, picked up after 4 rings
Appointments Booked	9 of 20 (45%)	2 of 20 (10%)	-77.8% drop: 5 phantoms + 2 timezone corrupted
Timezone Drift Bookings	0 (Unreported)	2 of 4 booked (50%)	C-08 & C-09 booked 5.5h late (UTC read as IST)
Phantom Bookings	0 (Unreported)	5 of 9 logged (55.6%)	C-02, C-03, C-05, C-10 (no book call), C-13 (Sunday error)
Vehicle Status Inquiries	Not tracked	8 of 20 calls (40%)	0 of 8 resolved: get_vehicle_status called 0 times
Dropped Transfers	0 (Unreported)	1 of 6 transfers (16.7%)	C-02 failed on no_answer; no callback logged

a. What Actually Happened This Week (< 250 words)

Weekly performance was severely overstated. The report claimed 9 of 20 calls (45%) were booked appointments. In reality, only 4 of 20 calls (C-01, C-08, C-09, C-18) executed book_appointment successfully. In 2 of those 4 (C-08, C-09), check_availability omitted start_local, causing the assistant to misread UTC as IST and book appointments 5.5 hours later than promised (15:30 IST vs 10:00 AM in C-08; 15:00 IST vs 9:30 AM in C-09). Only 2 of 20 calls (10%, C-01 and C-18) were booked accurately.

The other 5 of 9 claimed bookings (55.6%) were phantoms logged without an appointment: 4 callers only inquired about prices, pickup, or car status (C-02, C-03, C-05, C-10), and 1 (C-13) errored with WORKSHOP_CLOSED on a Sunday attempt.

Additionally, 8 of 20 callers (40%: C-02, C-04, C-07, C-11, C-14, C-16, C-17, C-20) had open job cards and asked about cars in the shop. The assistant called get_vehicle_status 0 of 8 times (0%), pitching unwanted service slots to existing workshop customers (C-02, C-07, C-11, C-16, C-20) or falsely claiming it had no status visibility (C-04, C-14, C-17).

Finally, 1 of 6 transfers failed due to no_answer (C-02) without logging a callback, and 1 caller was turned away during regular Friday hours (C-17).

b. One Call Traced End to End Through Its Tool Log

Trace of Call C-02 (Mon, 11:15 am | 1 min 48 sec):
1. 11:15:22: lookup_customer {"phone":"9XXXXXX604"} → returned {"customer_id":"C6120","open_job_cards":["JC-2274"]}.
2. Transcript: Caller asked: *"I gave my car on Saturday. Is it ready?"*
3. Where it went wrong: Tool call check_availability at 11:15:58 (and repeated at 11:16:40).
Why: Per tool reference, a caller inquiring about an existing car must receive get_vehicle_status(job_card="JC-2274"). Instead, the assistant ignored open_job_cards and called check_availability twice, pitching an unwanted booking to an existing customer.
4. 11:17:02: Caller objected; assistant invoked transfer_to_advisor {"reason":"status_query"}, which returned status: "no_answer" after 6 rings.
5. Assistant promised someone would get in touch but failed to invoke log_callback, dropping the caller.
6. 11:17:05: It invoked log_outcome {"outcome":"booked"}, falsely logging a booking without calling book_appointment.

c. Note to the Workshop Manager (< 80 words)

To: *Workshop Manager, Sparsh Motors Kothrud*
Subject: *Assistant Audit & Corrections Update*

Hi Team,

Our weekly report showed 9 booked appointments, but only 2 are valid. Two were booked 5.5 hours late due to a time zone issue, and five were general inquiries mistakenly logged as bookings.

Customers asking about cars already in the shop were also offered new bookings instead of status updates.

We are updating the assistant tonight to fix status inquiries, correct booking timestamps, and capture missed callbacks. We will review tomorrow morning's calls together.

d. Ticket for Engineering (Exactly 10 Lines)

e. One Fix You Would Make Tonight & Verification Tomorrow

Tonight's Fix: A runtime middleware guardrail and prompt patch:
1. **Routing Rule:** In system prompt: *"If lookup_customer returns open_job_cards and caller asks about their vehicle, immediately invoke get_vehicle_status. Never offer service booking to existing workshop customers."*
2. **API Guardrail:** Intercept log_outcome: if outcome is "booked", verify book_appointment returned status: "ok" for this session; if not, rewrite outcome to "no_action" or "callback".
3. **Timezone Fallback:** If check_availability omits start_local, compute start_local = UTC + 5:30.

Verification Tomorrow Morning (09:00 - 12:00 IST):
1. Verify 100% of calls with outcome "booked" have a matching appointment_id.
2. Confirm zero check_availability calls for callers with active open job cards.
3. Compare spoken times against confirmed_local to verify zero time drift.

f. Two Places the AI Gave an Answer and You Did Not Take It

1. The AI summary claimed 9 appointments were booked (45% booking rate) and that assistant performance was strong with reduced advisor load.

What I did instead: I audited tool logs against book_appointment. I rejected 7 of 9 bookings: 5 had no booking call or errored (C-02, C-03, C-05, C-10, C-13), and 2 were booked 5.5 hours off due to UTC drift (C-08, C-09). Real booking rate is 10% (2 of 20).

2. An AI copilot suggested that the assistant was successfully handling customer inquiries because call durations were low (under 3 minutes) and transfers were down.
What I did instead: I inspected transcripts for all 8 callers with open job cards. Durations were short because frustrated callers hung up after the assistant pitched unwanted service slots (C-02, C-07) or falsely claimed no status visibility (C-04, C-14, C-17). get_vehicle_status was called 0 of 8 times. Low duration reflected customer abandonment, not resolution.

g. Session Log & Prompt History

Prompt History & Analytical Trajectory:
• **Step 1 (Ingestion & Audit):** Built automated parser audit_stats.py and call_details.py to iterate all 20 calls across transcripts, tool invocations, and responses.
• **Step 2 (Invariant Formulation):** Formulated 5 deterministic operational invariants: Booking Integrity, Timezone Consistency, Active Job Card Routing, Transfer Escalation, and Site Capability.
• **Step 3 (Anomaly Isolation):** Isolated C-08 and C-09 as UTC format anomalies where start_local was omitted for non-periodic services; isolated C-02 as a triple cascading failure.
• **Step 4 (Production Engine Implementation):** Architected PulseWorkshop 24x7 with FastAPI, dynamic configuration (pulse_config.py), live WebSocket stream, and runtime middleware guardrail (pulse_proxy.py) preventing regressions in real time.
• **Step 5 (Automated Testing):** Authored run_tests.py verifying all 20 call verdicts, proxy corrections, and metric calculations in CI/CD.

Exhibit: Full 20-Call Comprehensive Diagnostic Audit Matrix

Call ID	Day & Time	Caller Intent	Reported Outcome	Audited Reality	Invariant Violations
C-01	Mon 09:42	Swift periodic servicing	Booked	Booked (Sat 10:00)	None (Clean)
C-02	Mon 11:15	Car readiness check	Booked	Phantom (Status inquiry)	PHANTOM_BOOKING, IGNORED_JC, DROPPED_TRANSFER
C-03	Mon 15:30	WagonR 30k cost query	Booked	Phantom (Comparing shops)	PHANTOM_BOOKING (No appointment)
C-04	Mon 17:05	Baleno status inquiry	Transferred	Transferred (Wait 41s)	IGNORED_JC (get_vehicle_status bypassed)
C-05	Tue 10:20	City periodic cost query	Booked	Phantom (Call back husband)	PHANTOM_BOOKING (No appointment)
C-06	Tue 12:40	Rework rattle complaint	Transferred	Transferred (Wait 63s)	None (Advisor escalation)
C-07	Tue 14:18	Car pickup readiness	No action	Abandoned inquiry	IGNORED_JC (Pitched booking to active car)
C-08	Tue 16:50	AC cooling diagnostic	Booked	Corrupted (5.5h Drift)	TIMEZONE_MISMATCH (Told 10:00, booked 15:30)
C-09	Wed 09:30	i20 brake inspection	Booked	Corrupted (5.5h Drift)	TIMEZONE_MISMATCH (Told 09:30, booked 15:00)
C-10	Wed 11:55	Pickup & drop charge	Booked	Phantom (Price inquiry)	PHANTOM_BOOKING (No appointment)
C-11	Wed 14:22	Ordered part status	Callback	Callback (CB-311)	IGNORED_JC (get_vehicle_status bypassed)
C-12	Wed 17:40	Wheel alignment billing	Transferred	Transferred (Wait 29s)	None (Advisor escalation)
C-13	Thu 10:10	Sunday service booking	Booked	Phantom (Errored call)	PHANTOM_BOOKING (Tool error WORKSHOP_CLOSED)
C-14	Thu 12:05	Nexon status inquiry	Callback	Callback (CB-314)	IGNORED_JC (get_vehicle_status bypassed)
C-15	Thu 15:45	Bodyshop insurance dent	No action	Dropped inquiry	UNCONFIGURED_SERVICE (Bodyshop not on site)
C-16	Fri 09:15	3rd repeat status call	Transferred	Transferred (Wait 88s)	IGNORED_JC (Pitched slot to 3rd time caller)
C-17	Fri 13:30	Claim survey status	No action	Turned away in hours	IGNORED_JC (False claim of outside hours)
C-18	Fri 16:12	Creta 40k servicing	Booked	Booked (Tue 11:00)	None (Clean)
C-19	Sat 10:35	Two wheeler showroom	No action	Wrong number	None (Clean)
C-20	Sat 12:50	Urgent status travel	Transferred	Transferred (Wait 52s)	IGNORED_JC (Pitched slot to active car)